# Theory Lectures on

# Manufacturing Process (NMES 101)

## Allowances

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Content prepared for theory lecture for 1st Year BTech level students for common course on Manufacturing Processes. Each lecture duration ≈ 45 min

# Pattern making for sand casting



**The five basic steps in sand castings:**

1. Pattern making
2. Core making (optional)
3. Molding
4. Melting of metal and pouring into cavity
5. Cooling, mold dismantling, and cleaning the object

**Pattern making:**

**What is the pattern:** Replica of the part (either exact or with some modification) to be cast and is used to prepare the mold cavity. It is the physical model of the casting used to make the mold.

**Pattern Material**: Usually wood. But can be wax, metal, plaster-of-paris, or plastic.

The mold is made by packing some readily formed aggregate material, such as molding sand, surrounding the pattern.

When the pattern is withdrawn, its imprint provides the mold cavity. This cavity is filled with metal to become the casting.

This is the metal object you need

This is the wooden pattern required

**Desirable properties of pattern material**

- Easily worked, shaped and joined
- Light in weight
- Strong, hard and durable
- Resistant to wear and abrasion
- Resistant to corrosion, and to chemical reactions
- Dimensionally stable and unaffected by variations in temperature and humidity
- Available at low cost

# Wood and alternative pattern materials



## ➢Benefits of wood as pattern material

1. Cheap
2. Easily available,
3. Light weight,
4. Easiness in surfacing,
5. Easiness in preserving (by shellac coating),
6. Easily workable,
7. Ease in joining,
8. Ease in fabrication

## ➢Disadvantages of wood as pattern material

1. Absorbs moisture and changes dimension,
2. Wear by sand abrasion,
3. Warp during forming,
4. Not suitable for rough use,
5. Must be properly dried/ seasoned,
6. Should be free from knots.

**Common woods**: Burma teak, pine wood, mahogany, Sal, Deodar, Shisham, Walnut, Apple tree

## Metal as pattern material:

1. Used for mass production
2. For maintaining closer dimensional tolerances on casting.
3. More life when compared to wooden patterns
4. Do not absorb moisture
5. Do not wrap, retain their shape
6. Greater resistance to abrasion
7. Accurate and smooth surface finish
8. More expensive

**Common metals**: Al alloys, Brass, while metals, Mg alloys, Steel, cast Iron for mass production

## Plastic as pattern material:

1. Light weight
2. Good formability
3. Do not absorb moisture
4. Good corrosion resistance

**Common plastics**: Epoxy resins with fillers, PU foam

## Plaster-of-Paris as pattern material

1. Complicated shapes can be cast
2. Has high compressive strength
3. Used for making small and intricate pattern and core boxes

Wax is also used in specialized applications such as investment casting. It provide good surface finish and high accuracy

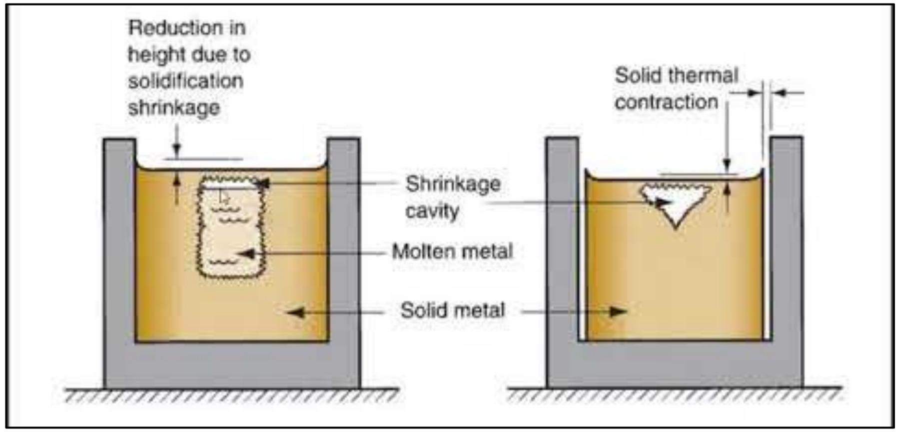
# Pattern allowances – Shrinkage allowance



In general, pattern has inverse shape of the original product that needs to be cast.

However, in practice, pattern is always made slightly larger than the final casting dimensions to take care of various practical limitations and distortions in dimensions. Such provisions (metallurgical and mechanical reasons) to provide dimension to the pattern is called "allowances". There are usually 5 types of allowances in foundry practices.

1. Shrinkage allowance
2. Draft allowance
3. Machining allowance
4. Distortion allowance
5. Rapping allowance




Desired product dimension ≠ Pattern dimension

1. In casting, there is phase transformation involved. Solid to liquid (heating) and once again liquid to solid (cooling).
2. Almost all metals shrink or contract volumetrically during cooling.
3. Our intension is to get the product of desired dimension. If the pattern dimension is same with desired dimension, then mold cavity will also have dimension same with desired dimension. Molten metal will initially fill the cavity; but during solidification, the metal will automatically contract. So the final solid product will have dimension slightly lower than the cavity or intended product.
4. Shrinkage allowance is given on the pattern to compensate for the contraction of the liquid metal on cooling.
5. For this, the dimensions of the pattern is made slightly oversize (Positive Allowance).
6. The shrinkage allowance will be different for different metals depending on the volumetric shrinkage of the concerned cast metal.
7. The shrinkage allowance is always added along the length.

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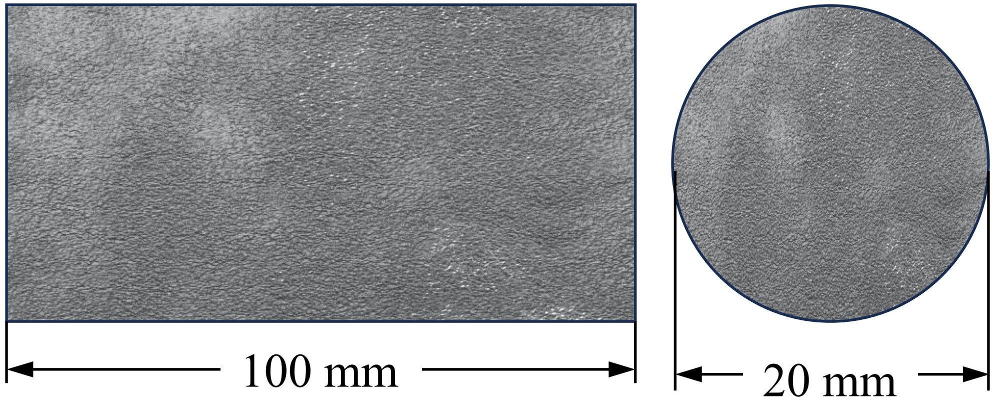
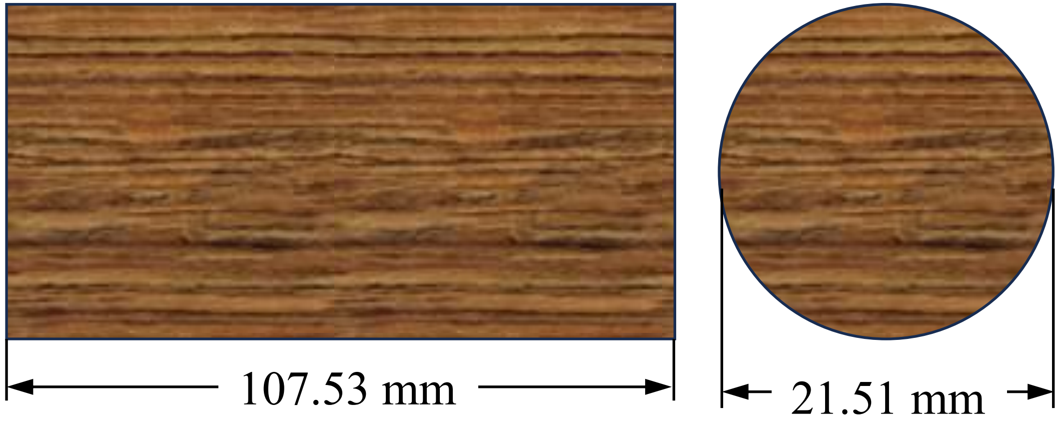
# Shrinkage allowance problem



**Problem**: An aluminum rod of 100 mm length and 20 mm diameter is to be made by sand casting. Aluminum contract about 7% during solidification. Calculate the desired dimension of the wooden pattern considering only shrinkage allowance




(i) This is the desired product




(ii) This should be the wooden pattern

**Shrinkage allowance** is always added along the exterior dimension. The pattern is always oversize to account for volumetric shrinkage.

We will always focus on desired length. This length is fixed. We need to achieve that even after contraction.

**First focus on length**

- The desired length is 100 mm
- The concerned metal shrinks by 7%
- So the pattern dimension has to be bigger than 100 mm
- Say, x is the pattern length. Therefore,
- $x - \frac{7x}{100} = 100$
- $x = 107.53$ mm
- This should be the pattern length to achieve 100 mm cast product even after shrinkage of the aluminum

**Now focus on diameter**

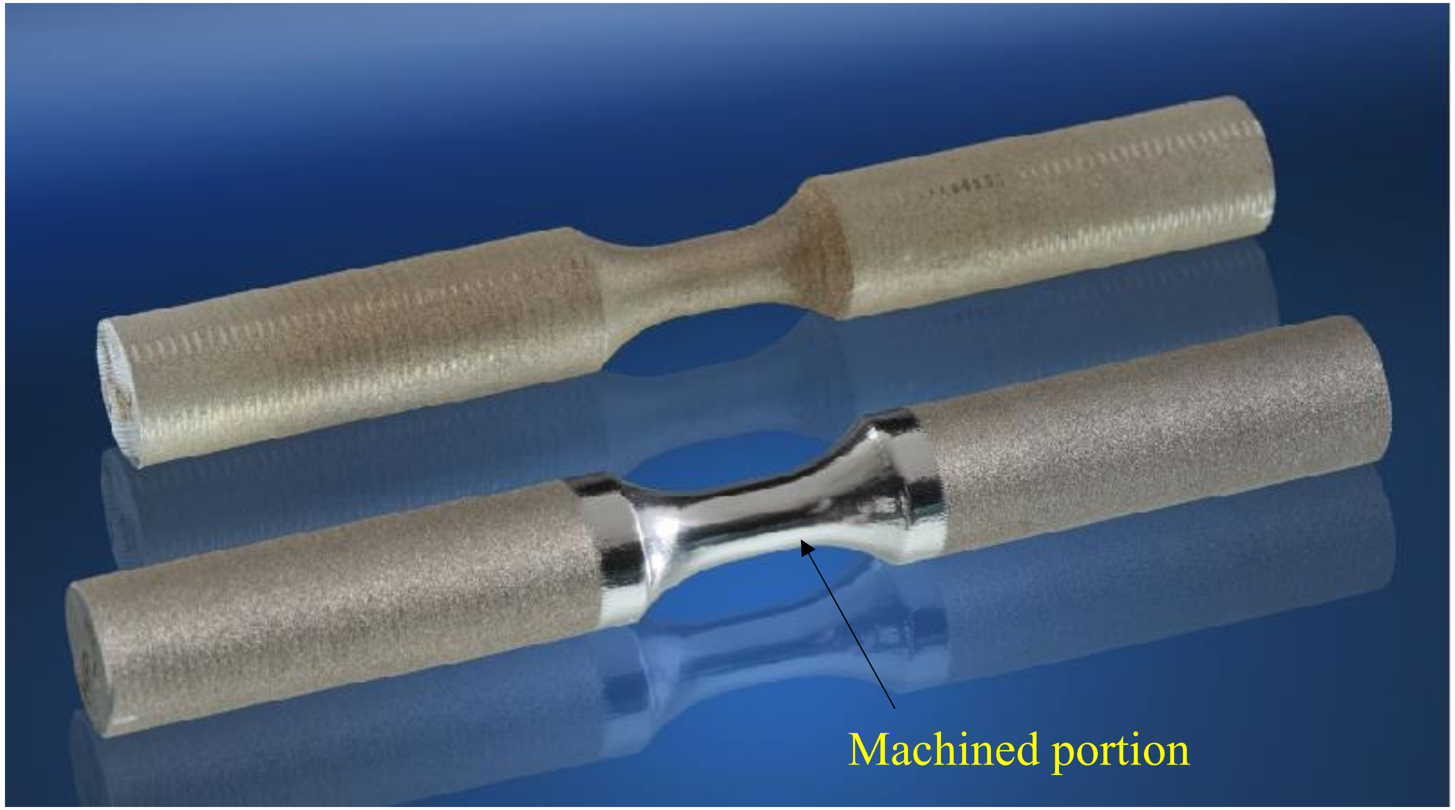
- The desired length is 20 mm and the concerned metal shrinks by 7%
- Say, y is the pattern diameter. Therefore,
- $y - \frac{7y}{100} = 20 \quad \Rightarrow y = 21.51$ mm

# Machining allowance



- **Machining** is a process for finishing the product.
- It basically removes a thin layer of material from the product surface to generate a smooth surface.

- The final metal object obtained by casting usually has very rough surface (because the molten metal interfaces with rough sand surface within the cavity. The same gets solidified resulting in rough surface generation).
- Thus, the final cast product is usually finished by machining to improve surface finish.
- As a result, a thin layer of metal is sacrificed from cast product to get desired smooth surface.
- Accordingly, the cast product must be slightly oversize so that the desired dimension is retained even after machining.
- This requires an oversize cavity or perhaps an oversize pattern.
- This is achieved by providing Machining Allowance to the pattern.

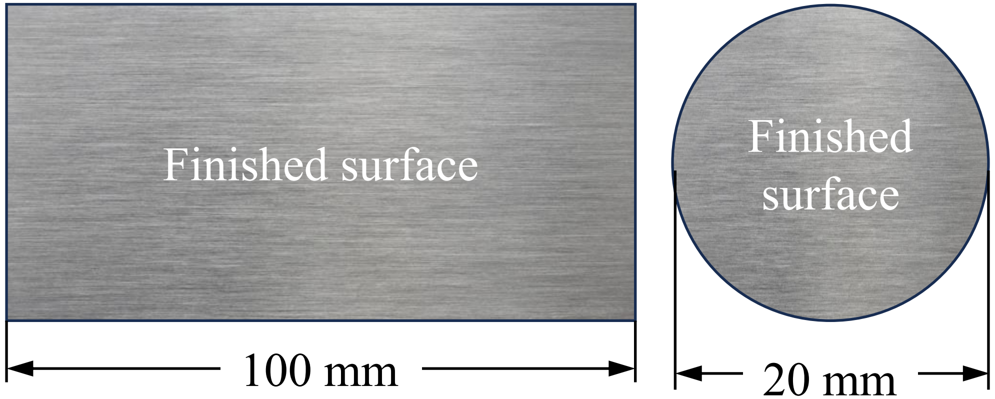



- Note that shrinkage allowance depends on the casting metal and its volumetric shrinkage amount.
- But machining allowance usually has fixed value. It depends on how much material you need to remove by machining to finish the surface.
- Usually it lies within 0.5 – 2.0 mm.
- This much thickness of material needs to be removed from surface by machining.

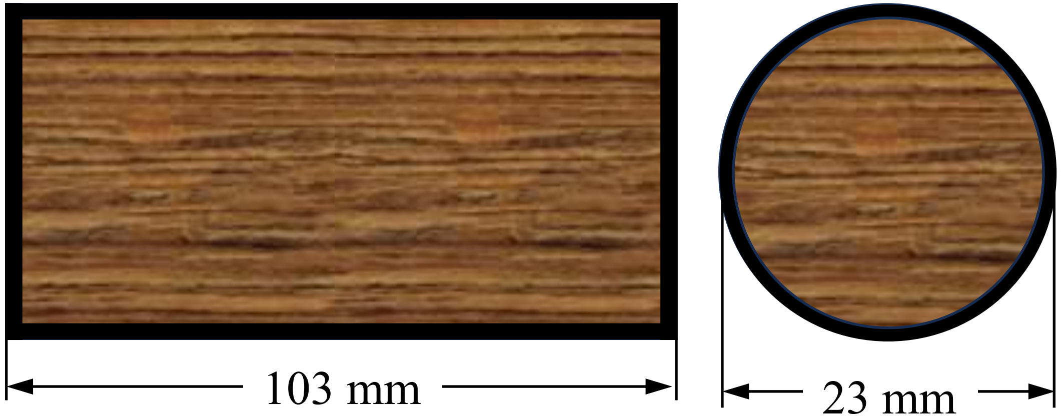
# Machining allowance problem



**Problem**: An aluminum rod of 100 mm length and 20 mm diameter is to be made by sand casting. The final cast product must be a finished one with roughness within 1.0 µm. Multi-stage machining need to be carried out to achieve this level of finished surface. During first stage machining, 1.0 mm should be removed from every side. In second stage, 0.5 mm should be removed from every side. Calculate the desired dimension of the wooden pattern considering **only machining allowance**.




(i) This is the desired product




(ii) This should be the wooden pattern

Pattern must be oversize than desired dimension so that material can be removed from cast surface to obtain finished product.

- It is multi-step machining process.
- 1.0 mm needs to be removed in first step. 0.5 mm in second step.
- Total removal = 1.5 mm from every side.

**First focus on length**
- Desired length = 100 mm
- Machining must be carried out in both the ends.
- Machining allowance = 1.5 × 2 = 3.0 mm
- Pattern length = 100 + 3 = 103 mm.

**Now focus on diameter**
- Desired diameter = 20 mm
- Machining must be carried out in entire circumference.
- Machining allowance = 1.5 mm (radial) => 3.0 mm (diametral)
- Pattern diameter = 20 + 3 = 23 mm.

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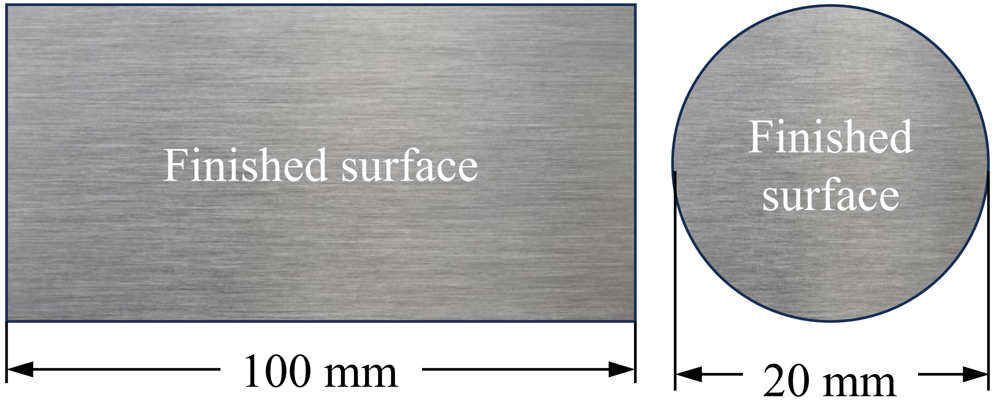
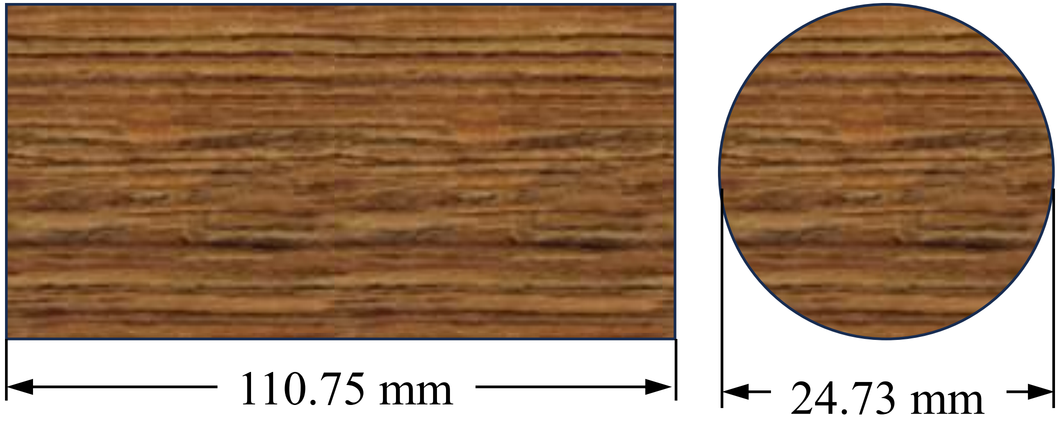
# Combined machining & shrinkage allowances problem



**Problem**: An aluminum rod of 100 mm length and 20 mm diameter is to be made by sand casting. The final cast product must be a finished one with roughness within 1.0 µm. Multi-stage machining need to be carried out to achieve this level of finished surface. During first stage machining, 1.0 mm should be removed from every side. In second stage, 0.5 mm should be removed from every side. Aluminum contract about 7% during solidification. Calculate the desired dimension of the wooden pattern considering both machining and shrinkage allowances.




(i) This is the desired product




(ii) This should be the wooden pattern

Pattern/Cavity => Metal shrinkage => Machining => Desired product

? <= (+)7% <= (+)1.5 mm <= 100 mm

**First focus on length**

- Desired length = 100 mm
- Machining allowance = 1.5 × 2 = 3.0 mm
- Length of pre-machined product = 100 + 3 = 103 mm
- Shrinkage allowance = 7%
- Now if final pattern length be x, then
- $x - \frac{7x}{100} = 103 \quad \Rightarrow x = 110.75$ mm

**Now focus on diameter**

- Desired diameter = 20 mm
- Machining allowance = 1.5 mm (radial) = 3.0 mm (diametral)
- Pre-machined product diameter = 20 + 3 = 23 mm
- Now if final pattern diameter be y, then
- $y - \frac{7y}{100} = 23 \quad \Rightarrow y = 24.73$ mm

# Practice set



**Problem 1**: A 100 mm long and 20 mm diameter cylindrical component is desired. To manufacture this, first sand casting process is used to make a cylinder having rough surfaces. This cast product is then machined in single-stage turning to enhance the surface finish by removing 1.0 mm thin layer of material from every side. Volumetric shrinkage of the concerned metal is known to be 7%. Calculate the desired dimension of the wooden pattern used in sand casting considering both machining and shrinkage allowances.

**Problem 2**: A steel rod of 150 mm long and 25 mm diameter is to be made by sand casting. After casting, multi-stage machining needs to be carried out to achieve the customer-specified surface finish level. During first stage machining, 0.75 mm should be removed from every side. In second stage machining, 0.25 mm should be removed from every side. Steel contract about 6% during solidification. Calculate the desired dimension of the wooden pattern considering both machining and shrinkage allowances.

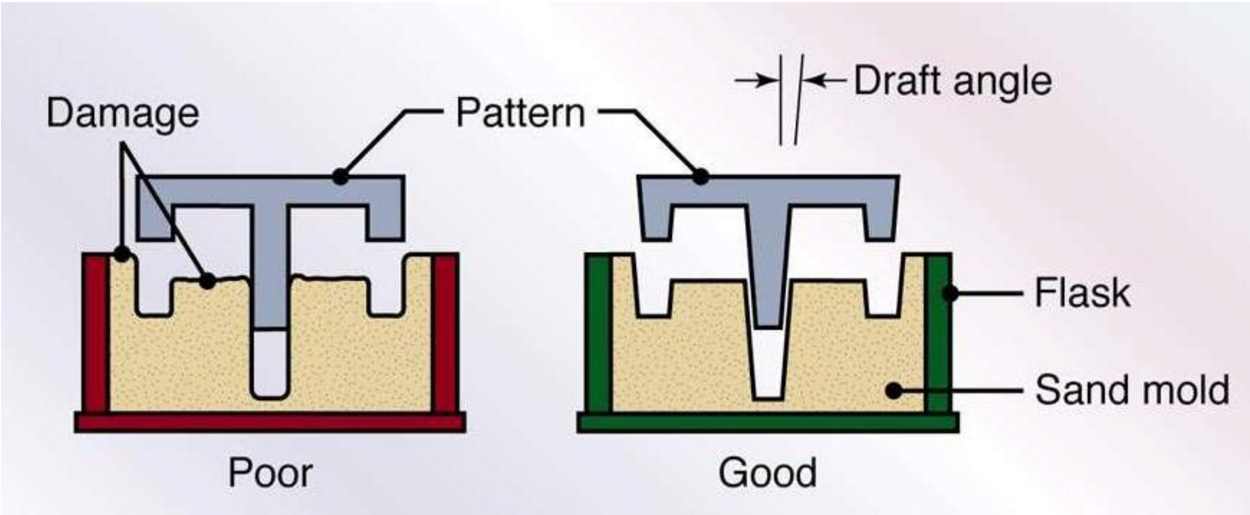
**Problem 3:** A cylindrical steel rod with smooth surface is to be manufactured by casting and machining. The final desired product dimension should be 140 mm in length and 46 mm in diameter. The wooden pattern has 150 mm length and 50 mm diameter. During solidification, the cast metal experiences uniform contraction of 6% in all the directions. Single stage machining is carried out to smoothen the surfaces of the cast product. Considering both machining and shrinkage allowances, calculate the thickness of material that should be removed from all directions during machining.

**Problem 4**: A cubic steel block having smooth surfaces needs to be manufactured by casting and machining. The side length of final desired cubic product should be 280 mm. A cubic wooden pattern is fabricated with 300 mm side length. During solidification, the cast metal experiences 6% volumetric shrinkage. Single stage machining is carried out to smoothen the surfaces of the cast product. Considering both machining and shrinkage allowances, calculate the thickness of material that should be removed from all directions during machining.

# Draft allowance and Shaking allowance

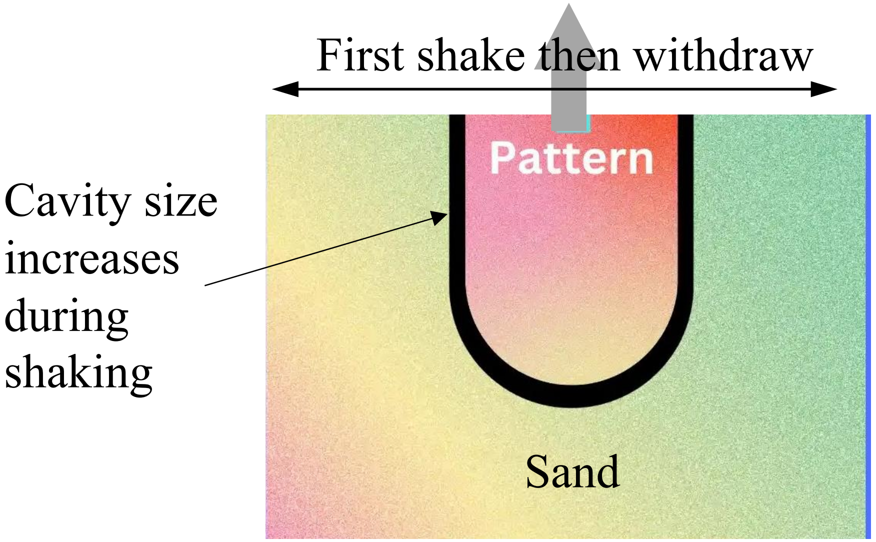


## Draft or Taper Allowance




- Wooden pattern must be taken out (drafting) from the sand mold at the final step of molding in order to have the cavity where metal can be poured.
- During this withdrawal of the pattern, there exists a high chance of breakage of the vertical sand walls.
- To minimize the chances of breakage of vertical sand walls, the vertical walls of the pattern are intentionally made slightly tapered. This is known as draft/taper allowance.
- It is provided to facilitate easy withdrawal of the pattern.
- Typically it ranges from 1 to 3 degree for wooden patterns.

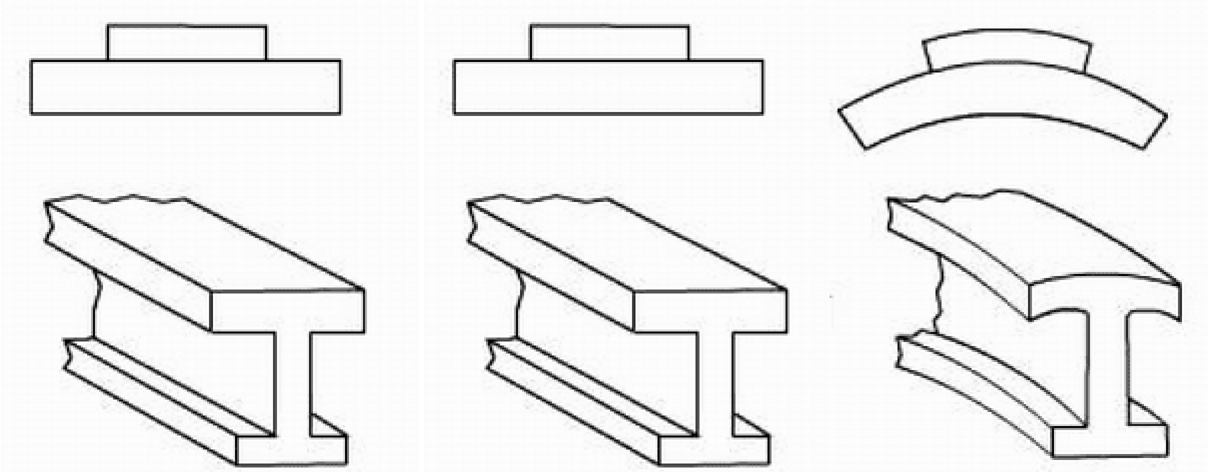
## Rapping or Shaking Allowance




- How can you easily withdraw a pin from a wall? By shaking it first. Right?
- Similarly, during withdrawal of the pattern, it is often shook (oscillated / waggled) for easy withdrawal. As a result, the sand walls get distorted and the cavity dimension increases slightly.
- In order to compensate for this increase, the pattern is initially made slightly smaller. So it is a negative allowance.
- The amount of this allowance is given based on previous experience (operator specific – usually a fixed value).

# Distortion Allowance

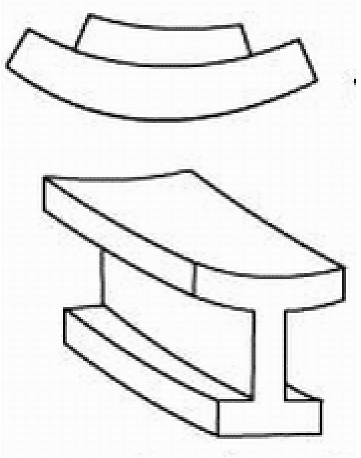




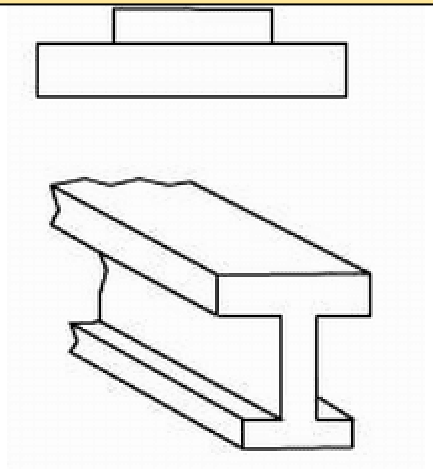
Desired shape Pattern shape Final cast shape

This distortion problem occurs during casting

To avoid this problem, intentionally give a cambered shape (inverse to the distortion) to the pattern so that after distortion the cast product attains the desired shape.



Pattern shape



Desired shape

- Casting having shapes such as C, U and large plate, can loose their shapes during solidification. The loss of shape is referred to as distortion.
- This distortion occurs due to the shrinkage stresses present during solidification.
- To take care of this, the pattern is given an allowance in the direction opposite to the expected distortion.
- This is referred to as distortion allowance.
- Therefore, the pattern is intentionally made cambered (instead of actual desired shape) so that the cast product attains desired shape after distortion.
- Experience is essential in addition to the design knowledge in arriving to this allowance.

# Videos related to pattern allowances



Videos related to pattern allowances

- https://www.youtube.com/watch?v=8EO25lnMj-I
- https://www.youtube.com/watch?v=4rzEGeiC7Zw
- https://youtu.be/H78qWl4sf54?t=1843